

Lab 2

1. Display the number of students their name is "Mohammed"

```
ties  Terminal 30 Apr 12:55 PM
alaa@lol:~$ sudo -i -u postgres
psql
[sudo] password for alaa:
postgres@lol:~$ psql
psql (14.17 (Ubuntu 14.17-0ubuntu0.22.04.1))
Type "help" for help.

postgres=# \list

[1]+  Stopped                  psql
postgres@lol:~$ \c advpostgres
c: command not found
postgres@lol:~$ psql
psql (14.17 (Ubuntu 14.17-0ubuntu0.22.04.1))
Type "help" for help.

postgres=# \c advpostgres
You are now connected to database "advpostgres" as user "postgres".
advpostgres=# SELECT COUNT(*) AS number_of_mohammed
FROM student
WHERE e_name LIKE 'Mohammed%';
 number_of_mohammed
-----
0
(1 row)

advpostgres=#
```

2. Display the number of males and females.

```
advpostgres=# SELECT gender, COUNT(*) AS count
FROM student
GROUP BY gender;
 gender | count
-----+-----
 Male   |    3
 Female |    2
(2 rows)

advpostgres=#
```

3. Display the repeated first names and their counts if higher

than 2.

```
advpostgres=# SELECT split_part(e_name, ' ', 1) AS first_name, COUNT(*) AS count
FROM student
GROUP BY first_name
HAVING COUNT(*) > 2;
 first_name | count
-----+-----
(0 rows)

advpostgres=#
```

4. Display all students and track name they belong to.

```
advpostgres=# SELECT s.e_name, s.email, t.track_name
FROM student s
LEFT JOIN track t ON s.track_id = t.id;
 e_name      | email              | track_name
-----+-----+-----
Ahmed Youssef | ahmed.youssef@gmail.com | Web Development
Sara Mahmoud  | sara.mahmoud@gmail.com  | Data Science
Mohamed Ali   | mohamed.ali@gmail.com   | Cybersecurity
Nourhan Adel  | nourhan.adel@gmail.com  | AI and Machine Learning
Khaled Saeed  | khaled.saeed@gmail.com  | Operating Systems
(5 rows)
```

5. Display all students except those who are in OS track.

```
advpostgres=# SELECT s.e_name, s.email, t.track_name
FROM student s
JOIN track t ON s.track_id = t.id
WHERE t.track_name <> 'Operating Systems';
 e_name      | email              | track_name
-----+-----+-----
Ahmed Youssef | ahmed.youssef@gmail.com | Web Development
Sara Mahmoud  | sara.mahmoud@gmail.com  | Data Science
Mohamed Ali   | mohamed.ali@gmail.com   | Cybersecurity
Nourhan Adel  | nourhan.adel@gmail.com  | AI and Machine Learning
(4 rows)
```

6. Create a view for student names with their subjects' names

which they will study.

```
advpostgres=# CREATE VIEW student_subjects_view AS
SELECT s.e_name AS student_name, sub.sub_name AS subject_name
FROM student s
JOIN stu_sub ss ON s.id = ss.stu_id
JOIN subject sub ON ss.sub_id = sub.id;
CREATE VIEW
advpostgres=#
```

```
advpostgres=# CREATE VIEW student_subjects_view AS
SELECT s.e_name AS student_name, sub.sub_name AS subject_name
FROM student s
JOIN stu_sub ss ON s.id = ss.stu_id
JOIN subject sub ON ss.sub_id = sub.id;
CREATE VIEW
advpostgres=# SELECT * FROM student_subjects_view;
 student_name | subject_name
-----+-----
Ahmed Youssef | Databases
Sara Mahmoud  | Networks
Mohamed Ali   | Programming
Nourhan Adel  | Machine Learning
Khaled Saeed  | Operating Systems
(5 rows)
```

7. Create a view for tracks names and the subjects which belong to it.

```
advpostgres=# CREATE VIEW track_subjects_view AS
SELECT t.track_name, sub.sub_name
FROM track t
JOIN track_sub ts ON t.id = ts.track_id
JOIN subject sub ON ts.sub_id = sub.id;
CREATE VIEW
advpostgres=#
```

```
advpostgres=# CREATE VIEW track_subjects_view AS
SELECT t.track_name, sub.sub_name
FROM track t
JOIN track_sub ts ON t.id = ts.track_id
JOIN subject sub ON ts.sub_id = sub.id;
CREATE VIEW
advpostgres=# SELECT * FROM track_subjects_view;
 track_name | sub_name
-----+-----
Web Development | Databases
Data Science   | Networks
Cybersecurity  | Programming
AI and Machine Learning | Machine Learning
Operating Systems | Operating Systems
(5 rows)
```

```
advpostgres=# SELECT * FROM track_subjects_view;
      track_name      |      sub_name
-----+-----
Web Development      | Databases
Data Science         | Networks
Cybersecurity        | Programming
AI and Machine Learning | Machine Learning
Operating Systems    | Operating Systems
(5 rows)
```

```
advpostgres=# SELECT * FROM track_subjects_view;
      track_name      |      sub_name
-----+-----
Web Development      | Databases
Data Science         | Networks
Cybersecurity        | Programming
AI and Machine Learning | Machine Learning
Operating Systems    | Operating Systems
(5 rows)
```